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Cylinder Head Lifting Jib

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

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1.0 Introduction

Cylinder heads from large mining trucks, weighing approximately 50 kilograms each, are currently lifted manually from the engine, carried upwards out of the engine bay to the deck and are then carried down the stairs to ground level.

Cylinder head components need to be carefully positioned during removal and assembly. Furthermore, these components need to be maneuvered over the top of the engine for removal from the engine compartment. There is significant potential for manual handling related incidents and injuries when performing these tasks.

The following best practice outlines a means of supporting such components by implementing a fit for purpose lifting jib that can be used inside the engine compartment during servicing.

2.0 Best Practice Description

A lifting jib can be fabricated to attach to the engine step mounting points above the exhaust manifold. Figure 1 shows the concept of a simple jib fabricated on site for lifting cylinder heads during servicing.

Figure 1. Fabricated lifting jib with roller block

Figure 2. Jib supported by engine step mounting

Figure 2 and 3 shows the concept lifting jib that can be used both on workshop and field servicing. A jib such as this must be designed in a way so that it can be lifted into the engine compartment and installed easily. Both the boom and platform design shown in Figure 4 weigh less than 20kg as separate components. This allows the technician to carry the components up to the engine compartment for use.

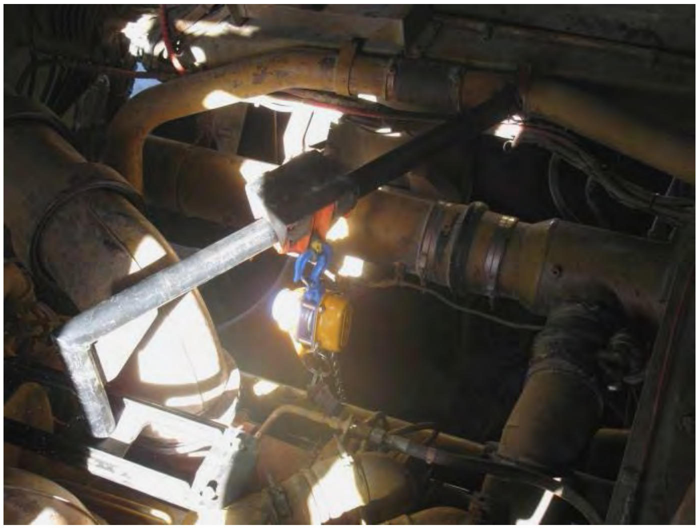

Figure 3. Concept lifting jib used in service

Figure 4. Lifting jib boom and platform

Before any fabricated lifting device can be used, the mine must have the device certified by a licensed lifting device inspector. This device must be put on a yearly (or more frequent) inspection with the other lifting devices on site.

3.0 Implementation Steps

- A portable lifting jib can be fabricated to suit anchor points on engine components such as the platform step anchors shown in figure 2.
- Each component must be manufactured and tested to relevant crane and lifting equipment standards and have a specified load rating. The jib must be “certified” by a licensed lifting device inspector.
- Lifting jib anchor points must place undue stress onto its supporting structures e.g. engine lift points and fasteners.

4.0 Benefits

1. Cylinder heads do not have to be removed and lifted manually from the engine compartment.
2. Cylinder heads can be positioned and supported correctly during assembly.
3. Swing boom allows cylinder heads to be maneuvered over other engine components.

5.0 Resources Required

The design of the lifting equipment is customized and price and availability will be dependant on size, arrangement and supplier.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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